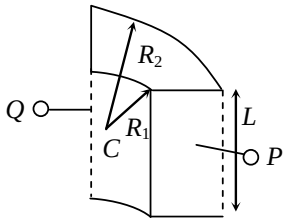
PHYSICS

SECTION – I
Single Correct Choice Type

This section contains 6 single correct choice type questions. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONLY ONE** is correct.

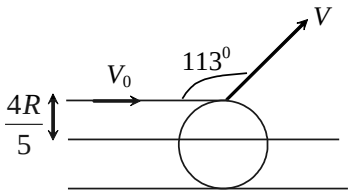
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1. A resistor is formed in the shape of a hollow, quarter cylinder from a material of resistivity ρ. The length of cylinder is L, inner and outer radii are R_1 and R_2 respectively. The resistance of this resistor between the shown terminals P and Q is



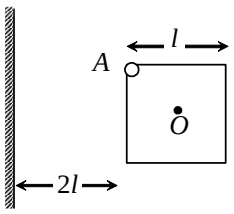
(a) $\dfrac{2\pi\rho L}{\left(R_2^2 - R_1^2\right)}$ (b) $\dfrac{2\rho\pi}{L\ln\left(\dfrac{R_2}{R_1}\right)}$ (c) $\dfrac{\pi\rho}{2L\ln\left(\dfrac{R_2}{R_1}\right)}$ (d) $\dfrac{\rho\pi R_1}{L(R_2 - R_1)}$

2. A smooth ball of steel is moving in horizontal direction with the speed of $V_0 = 5\sqrt{gR}$ where R is the radius of a smooth sphere fixed on the horizontal surface as shown in figure. The ball collides with the sphere and moves with changed velocity V along the line shown in figure. Then which of the following statements is correct?



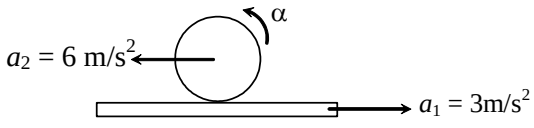
(a) $V = V_0$ (b) $4V = 5\,V_0$ (c) $8V_0 = 5\sqrt{3}V$ (d) $5V_0 = 8\sqrt{3}V$

3. A uniform wooden square board of side l is kept on a frictionless horizontal surface such that its distance from a vertical wall is $2l$ as shown in figure. A man having mass equal to the board is standing on the corner A of the board. If he moves to the centre O of the board then the distance of the centre of the board from the wall will become



(a) $\dfrac{\left(5\sqrt{2}-1\right)}{2\sqrt{2}}l$ (b) $\dfrac{\left(5\sqrt{2}\right)}{4}l$ (c) $\dfrac{5l}{2}$ (d) $\dfrac{9l}{4}$

4. In the following figure, a sphere of radius 3m is in pure rolling on the plank. The accelerations of the sphere and the plank are indicated. The value of angular acceleration α is

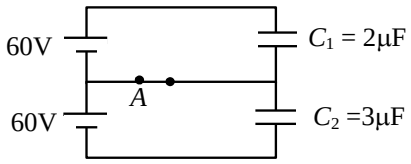


(a) 3 rad/s^2 (b) 6 rad/s^2

(c) 3 rad/s^2 (opposite to the direction shown in figure)

(d) 1 rad/s^2

5. Initially the switch *A* is open in the adjacent figure. Find the charge which flows through switch *A* after it is shorted.



(a) 24 μC (b) 36 μC

(c) – 36 μC (d) 60 μC

6. Standing waves are set up in a sting of length 240 cm clamped horizontally at both ends. The separation between any two consecutive points where displacement amplitude is $3\sqrt{2}$ cm is 20 cm. Standing waves were set by two traveling waves of equal amplitude of 3 cm. The overtone in which the string is vibrating will be

(a) 2nd (b) 3rd (c) 4th (d) 5th

SECTION – II
Multiple Correct Choice Type

This section contains 5 multiple correct choice type questions. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONE OR MORE** is/ are correct.

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7. A long plank of mass M = 20 kg can move on horizontal frictionless surface freely. Initially the plank is at rest and an object of mass m = 5kg is given a velocity of 25 m/s and placed on the floor of the plank. Co-efficient of friction between object and floor of the plank is $\frac{5}{8}$. After having some relative movement w.r.t. the plank, the object finally starts moving with the plank in one piece. Then

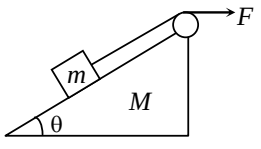
(a) the total work done by friction is –1250 J

(b) the minimum length of the plank should be 40 metres

(c) final velocity of plank is 10 m/s

(d) for time t = 3.2 seconds, the object is having relative motion w.r.t. plank

8. There is no friction anywhere in the system shown. The pulley is light and the wedge is free to move on the friction less surface. A horizontal force F is applied on the system in such a way that m does not slide on M or both move together with some common and constant acceleration. Given $M > m\sqrt{2}$



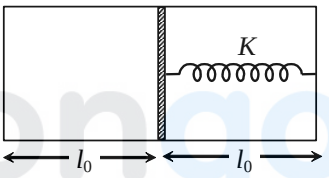
(a) Pseudo force acting on m as seen from frame of M is equal to $\frac{MF}{M+m}$ and is less than $mg \sin\theta$.

(b) Pseudo force acting on M as seen from the frame of m is greater than $\frac{mF}{m+M}$ and less than $mg \sin\theta$.

(c) Normal force between ground and M is greater than both $\frac{mF}{m+M}$ and mg sin θ

(d) Normal force (for θ = 45°) between m and M is equal to $\frac{mF}{m+M}$

9. A diathermic piston of mass M, cross section area A separate the volume inside a horizontal adiabatic cylinder of length $2l_0$ in two equal parts. Each chamber contains an ideal gas and pressure on each side is P. The piston can move without friction and is attached with a spring of spring constant K as shown. Initially the spring is non deformed. The piston is given a small displacement x towards left. Then



(a) the pressure in left chamber increases

(b) the pressure in right chamber decreases

(c) the piston oscillates with time period $2\pi\sqrt{\frac{Ml_0}{2PA+l_0K}}$

(d) the piston oscillates with time period $2\pi\sqrt{\frac{2Ml_0}{PA+l_0K}}$

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10. A charge particle of charge q, mass m is projected with velocity v in a region of electric and magnetic fields E and B respectively, such that v, E and B are mutually perpendicular to each other and $v = E/B$. Then (neglect gravity)

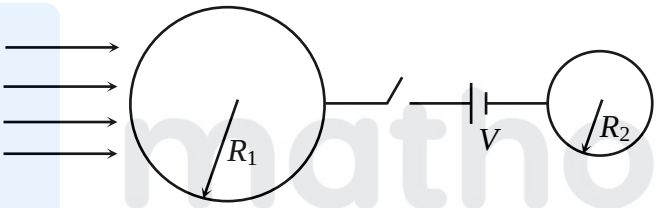
(a) If E is switched off, then particle will follow a circular path

(b) If B is switched off, then particle will follow a parabolic path

(c) If E is switched off, for a time of $\frac{\pi m}{4qB}$ and then switched on, the path of particle will not be a straight line

(d) If E is switched off, for a time of $\frac{2\pi m}{qB}$ and then switched on, the path of particle will be a straight line

11. Two conducting uncharged spheres of radius R_1 and R_2 ($R_1 > R_3$) are connected to a battery with a switch as shown. Light rays of frequency of f are incident on the bigger sphere and simultaneously the switch is closed. Work function of bigger sphere is ϕ. After some time the charge on bigger sphere becomes q_1, and on smaller sphere becomes $-q_2$ remains constant there after. Then



(a) $\frac{q_1}{4\pi\varepsilon_0 R_1} + \frac{q_2}{4\pi\varepsilon_0 R_2} = V$

(b) The number of electrons emitted by larger sphere is $\frac{q_1 - q_2}{e}$ when e is the charge one electron

(c) $\frac{eq_1}{4\pi\varepsilon_0 R_1} = hf - \phi$

(d) The number of electrons emitted by larger sphere is independent of the potential of the battery

SECTION – III

Reasoning Type

This section contains 5 reasoning type questions. Each question contains Statement-1 and Statement-2. Each question has 4 choices (a), (b), (c) and (d), out of which **ONLY ONE** is correct.

Directions: Read the following questions and choose

(A) **Statement-1** is True, **Statement-2** is True; **Statement-2** is a correct explanation for **Statement-1**.

(B) **Statement-1** is True, **Statement-2** is True; **Statement-2** is not a correct explanation for **Statement-1**.

(C) **Statement-1** is True, **Statement-2** is False.

(D) **Statement-1** is False, **Statement-2** is True.

12. **Statement-1**: Two blocks A and B of mass M and $2M$ respectively are connected by a massless spring of force constant K. They are stretched by an amount of x and then released on a smooth horizontal table. The relative velocity of the blocks when spring comes to its natural length is $x\sqrt{\frac{3K}{2M}}$.

Statement-2: When there is no external force acting on a system, its momentum and mechanical energy is always conserved.

(a) A (b) B (c) C (d) D

13. **Statement-1**: The time period of simple oscillation of a mass supported by a spring is T. The spring is cut into 4 equal parts and the body is again suspended by one of the pieces. The new time period of simple oscillation will be $\frac{T}{2}$.

Statement-2: The spring constant of a spring is inversely proportional to the length of the spring.

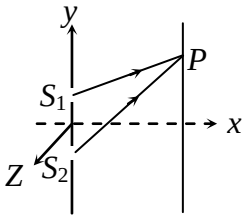
(a) A (b) B (c) C (d) D

14. **Statement-1**: If in an A.C. circuit, the applied voltage is in phase with the current flowing in the circuit, then the circuit may have only resistance in it.

Statement-2: If there is no phase difference between current and applied voltage, the power factor of the circuit is unity.

(a) A (b) B (c) C (d) D

15. **Statement-1**: In a young's double slit experiment set up shown in the figure, screen is perpendicular to x-axis. The fringes will be hyperbolic in shape and central bright fringe will be a straight line.



Statement-2: Each fringe will be the locus of a point for which $S_2P - S_1P$ will be a constant

(a) A (b) B (c) C (d) D

16. **Statement-1**: Two particles are projected towards each other from a very large distance along the line joining them on a smooth frictionless surface. Their speeds may decrease as they approach each other.

Statement-2: The net force on the system of the above two particles will be zero.

(a) A (b) B (c) C (d) D

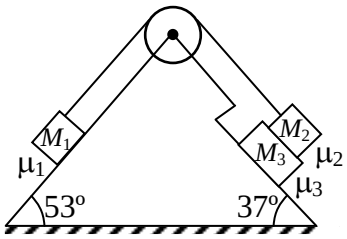
SECTION–IV
Matrix–Match Type

This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled 1, 2, 3, 4 and 5. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II**. The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:

If the correct matches are A – 1, 4 and 5; B – 2 and 3; C – 1 and 2; and D – 4 and 5; then the correct darkening of bubbles will look like the following.

	1	2	3	4	5
A	●	○	○	●	●
B	○	●	●	○	○
C	●	●	○	○	○
D	○	○	○	●	●

1. In the given arrangement there are three boxes with masses M_1, M_2 and M_3, 'μ_1, μ_2, μ_3' are the co-efficient of friction between M_1 and inclined plane, between M_2 & M_3, between M_3 & inclined plane respectively. Column I gives the values of parameter $M_1, M_2, M_3, \mu_1, \mu_2$ and μ_3 respectively and column II gives certain aspects of the system corresponding to the values given in column I.



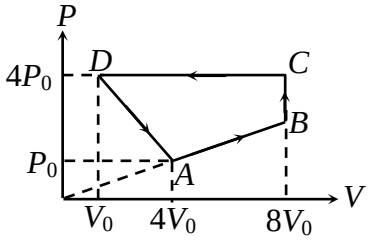
f_1 = friction force between M_1 and inclined plane

f_2 = friction force between M_2 & M_3

f_3 = friction force between M_3 and inclined plane (sin 37^0 – 3/5, sin 53^0 = 4/5)

Column I	Column II
(A) M_1 = 5kg, M_2 = 3kg, M_3 = 2kg, $\mu_1 = \mu_2 = \mu_3 = 0$	**1.** acceleration of M_1 and M_2 is same
(B) M_1 = 5kg, M_2 = 3kg, M_3 = 2kg, $\mu_1 = \mu_3 = 0, \mu_2 = 3/4$	**2.** acceleration of M_2 and M_3 is same
(C) $M_1 = M_2$ = 4kg, M_3 = 2kg $\mu_1 = 1/2, \mu_2 = 3/4, \mu_3 = 0$	**3.** $f_1 \geq f_2$
(D) $M_1 = M_2$ = 4kg M_3 = 2kg $\mu_1 = \mu_3 = 1/2, \mu_2 = 3/4$	**4.** $f_1 < f_2$
	5. $f_2 \leq f_3$

2. $\frac{3}{25}$ moles of an ideal monatomic gas undergoes a process as shown in figure. ΔU, Q and W represents the change in internal energy, amount of heat exchange and pressure-volume work respectively for the gas either for single process or for complete cycle. ($R = \frac{25}{3}$ SI unit)

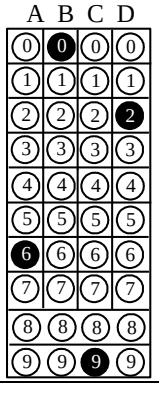


Column I		Column II	
(A)	In process AB	**1.**	$Q = 0$ or Q = +ve
(B)	In process DA	**2.**	$\Delta U = 0$ or ΔU = –ve
(C)	In cyclic process $ABCD$	**3.**	ΔU = +ve
(D)	In process CD	**4.**	W = +ve or $W = 0$
		5.	W = –ve

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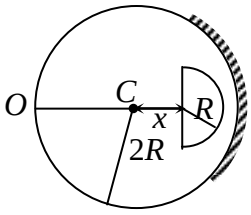
SECTION–V
Integer Answer Type

This section contains 4 questions. The answer to each of the questions is a **single digit integer,** ranging from 0 to 9. The appropriate bubbles below the respective question numbers in the ORS have to be darkened. For example, if the correct answers to question numbers A, B, C and D (say) are 6, 0, 9 and 2, respectively, then the correct darkening of bubbles will look like the following:



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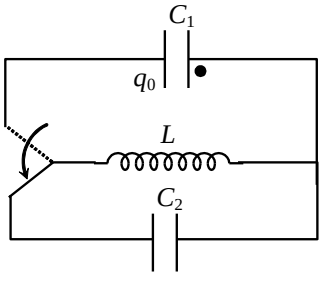
1. A hemispherical cavity of radius R is carved out from a sphere (μ = 1.5) of radius $2R$ such that principal axis of cavity coincides with that of sphere. One side of sphere is silvered as show in figure. Find the value of x (in cm) for which the image of an object at O is formed at O it self. (Take R = 8 cm and $\sqrt{145}$ =12)



Space for work

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2. At time $t = 0$ a charge q_0 is given at the plate of capacitor C_1 (C_1 = 400 μF) forming an LC circuit with inductor L = 100 H. At $t = \frac{\pi}{10}$ seconds connection with the capacitor C_1 is broken and another capacitor C_2 is joined with the inductor. At time $t = \frac{11n\pi}{200}$ the energy stored in capacitor C_2 is 36% of the total energy available finally. If C_2 = 100 μF find the value of n, $\left(\sin\left(\frac{\pi}{10}\right) = \frac{3}{10}\right)$



Space for work

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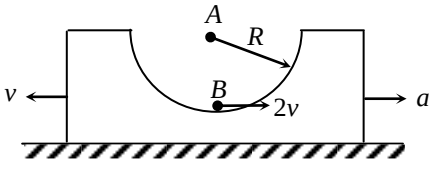
3. A non-conducting uniformly charged sphere having charge ‘96 columbs’ and radius ‘10m’ is placed with its center at origin, at a distance of ‘5m’ from the origin there is a large imaginary sheet placed, the flux of electric field through the intersection area of the sheet and the sphere is '$x/\in_0$'. Calculate the value of x?

Space for work

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4. A bead kept at the bottom of a wedge moves towards right with a velocity $2v$. If the wedge moves towards left with a velocity v while accelerating towards right with an acceleration a as shown then the magnitude of acceleration of the bead is x units. Find x if

$v = 3$ and $R = 9$ units.



Space for work

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